Chang He Aaron

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EDUCATION

University of New South Wales (UNSW) — Sydney, Australia
Bachelor of Science in Mathematics, UNSW 3956 — Term 3, 2024 – Present

- **Expected graduation:** Term 3, 2028 (Honours).
- **Planned transfer:** Bachelor of Computer Science, UNSW 3779.
- **Academic standing:** WAM **85**; admitted to the UNSW Talented Students Program (TSP) in 2025.

RESEARCH INTERESTS

Quantum computing: quantum algorithms and quantum error correction, including fault-tolerant computation, stabilizer/CSS codes, and algebraic methods in finite-dimensional quantum systems.
Mathematics: algebraic methods for isomorphism problems, multilinear algebra and tensor spaces, group actions, canonical forms, and related representation-theoretic tools.
Theoretical computer science: computational complexity and algorithms, especially graph and tensor isomorphism, algebraic reductions, and structural problems at the interface of algebra and computation.

RESEARCH EXPERIENCE

Independent Study, UTS — Sydney, Australia
Quantum Computing, Algebraic Algorithms, and Isomorphism Problems with Youming Qiao — Term 2, 2025 – Present

- Conducting supervised reading and research preparation in quantum computation, complexity theory, and algebraic methods for isomorphism problems.
- Working on graph isomorphism and tensor isomorphism problems, with emphasis on algebraic formulations, reductions, and invariants for equivalence testing.
- Preparing a manuscript on connections between graph isomorphism and tensor isomorphism.

Talented Students Program, UNSW — Sydney, Australia
Academic Reading in Large Language Models with Quoc Le Gia — Term 1, 2025

- Completed supervised academic reading through the UNSW Talented Students Program, focusing on selected literature in large language models.
- Developed practice in reading research papers, extracting technical assumptions, and presenting mathematical and algorithmic ideas clearly.